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Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta k v = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \overset{electromagnetic}{=} \ldots = \sqrt{\frac{1}{LC}} \overset{QM}{=} \frac{1}{2}\frac{p^2}{m}\frac{1}{\hbar} \tag{1}$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v}\frac{1}{\lambda} = \frac{\omega}{k}\frac{1}{\lambda} = \frac{\omega}{2\pi} \tag{2}$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{\hbar}\omega\mathcal{T} + \nu_0 \tag{3}$$

$$\begin{gathered} e = \text{electron charge} \qquad V_0 = \text{Opposing Potential } (I = 0) \\ \text{Energy Level: } i = [1, n] \in \mathbb{N} \end{gathered} \tag{4}$$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \tag{5}$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurance of an event/state by space distance and time periodicity as well as movement speed and reach length, $\quad v^2 = v_{ph} v_{gr}, \quad \omega t = 2\pi\nu t = \lambda \neq 0 \quad :$

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \tag{6}$$

Rest state of source charge

Statics localization of charge. Only spacial demependeny, no time dependency.

Energy transport ... Information transmission ...

$v = 0 \qquad \dot{v} = 0$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v = const. \ll c \qquad \dot{v} = 0$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \approx c \qquad \dot{v} = 0$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$v \neq 0 \qquad \dot{v} \neq 0$

Movement in Vacuum

$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{c}{v_{Ph}} = 1 \qquad 2\pi = k\lambda \qquad v_{ph} = c \qquad \omega = kc$

$\varepsilon_0\mu_0 = \frac{1}{c^2}$

Continuity: $\nabla \cdot j = i\omega\varrho \qquad i\omega Q = 0$

Movement in Matter

$\eta = \frac{\tan\alpha}{\tan\beta} = \frac{c}{v_{Ph}} = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \neq 1 \qquad v_{ph} = \pm\sqrt{\frac{1}{\varepsilon\mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \qquad \omega = kv_{ph} = kc\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}} = kc\frac{\tan\beta}{\tan\alpha} = k\frac{c}{\eta}$

$\varepsilon\mu = \left(\frac{\eta}{c}\right)^2$

Dispersion: $k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$

Near Field

$r_0 \ll r \ll \lambda \qquad kr \ll 2\pi \qquad \frac{w_m}{w_e} \ll 1$

r is small, fast movement

Far Field

$r \gg \lambda \qquad kr \gg 1 \qquad r \gg kr_0^2 \ll 1 \qquad \frac{w_m}{w_e} = 1$

r is big, slow movement

High Frequency

$\omega \gg \frac{2}{\mu\sigma r_0^2}$

λ is big, ν frequency i small, short time period

Low Frequency

$\omega \ll \frac{2}{\mu\sigma r_0^2}$

λ is small, ν frequency i big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global: ....

Local: ....

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1}\mathcal{D} = \mathcal{E}^{-1}(\mathcal{P} + \varepsilon_0\mathcal{B})$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1}\mathcal{H} = \mathcal{B}^{-1}\left(\frac{1}{\mu_0}\mathcal{B} - \mathcal{M}\right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1+\omega^2 t^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2}\varepsilon_0\mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta\frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan\alpha}{\tan\beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$